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Lab task # 3

Object Oriented Programming

Question: 1

Write a C++ program that asks the user to enter the number of rows and columns for a 2D integer array and then inputs all the elements of the matrix from the user. After storing the values, the program should check each element of the 2D array and determine whether it is a odd number or not. The program should count how many odd numbers are present in the matrix.

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 int main() 4 { 5 int rows, column; 6 cout << "Enter number of rows: "; 7 cin >> rows; 8 cout << "Enter number of columns: "; 9 cin >> column; 10 int arr1[rows][column]; 11 // Input values 12 for (int i = 0; i < rows; i++) 13 { 14 for (int j = 0; j < column; j++) 15 { 16 cout << "Enter value for row " << i + 1</pre>	<pre>Enter number of rows: 2 Enter number of columns: 3 Enter value for row 1 column 1: 1 Enter value for row 1 column 2: 4 Enter value for row 1 column 3: 2 Enter value for row 2 column 1: 6 Enter value for row 2 column 2: 8 Enter value for row 2 column 3: 2 Odd numbers are: 1 === Code Execution Successful ===</pre>

main.cpp	Output
<pre>18 cin >> arr1[i][j]; 19 } 20 } 21 // Print odd numbers 22 cout << "\nOdd numbers are:\n"; 23 for (int i = 0; i < rows; i++){ 24 for (int j = 0; j < column; j++) 25 { 26 if (arr1[i][j] % 2 != 0) 27 { 28 cout << arr1[i][j] << " "; 29 } 30 } 31 cout << endl; 32 } 33 return 0; 34 }</pre>	<pre>Enter number of rows: 2 Enter number of columns: 3 Enter value for row 1 column 1: 1 Enter value for row 1 column 2: 4 Enter value for row 1 column 3: 2 Enter value for row 2 column 1: 6 Enter value for row 2 column 2: 8 Enter value for row 2 column 3: 2 Odd numbers are: 1 === Code Execution Successful ===</pre>

Question: 2

Write a C++ program that asks the user to enter the number of rows and columns for a 2D integer array and then inputs all the elements of the matrix from the user. After storing the values, the program should check each element of the 2D array and determine whether it is a prime number or not. The program should count how many prime numbers are present in the matrix and display the total prime.

Solution:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 bool isPrime(int num) 5 { 6 if (num <= 1) 7 return false; 8 9 for (int i = 2; i <= num / 2; i++) 10 { 11 if (num % i == 0) 12 return false; 13 } 14 return true; 15 } 16 17 int main() 18 {</pre>		<pre>Enter number of rows: 3 Enter number of columns: 3 Enter value for row 1 column 1: 2 Enter value for row 1 column 2: 3 Enter value for row 1 column 3: 5 Enter value for row 2 column 1: 4 Enter value for row 2 column 2: 6 Enter value for row 2 column 3: 7 Enter value for row 3 column 1: 9 Enter value for row 3 column 2: 8 Enter value for row 3 column 3: 0 Prime numbers are: 2 3 5 7</pre>

main.cpp	Run	Output
<pre>19 int rows, column; 20 21 cout << "Enter number of rows: "; 22 cin >> rows; 23 24 cout << "Enter number of columns: "; 25 cin >> column; 26 27 int arr1[rows][column]; 28 29 // Input values 30 for (int i = 0; i < rows; i++) 31 { 32 for (int j = 0; j < column; j++) 33 { 34 cout << "Enter value for row " << i + 1 35 << " column " << j + 1 << ": "; 36 cin >> arr1[i][j];</pre>		<pre>Enter number of columns: 3 Enter value for row 1 column 1: 2 Enter value for row 1 column 2: 3 Enter value for row 1 column 3: 5 Enter value for row 2 column 1: 4 Enter value for row 2 column 2: 6 Enter value for row 2 column 3: 7 Enter value for row 3 column 1: 9 Enter value for row 3 column 2: 8 Enter value for row 3 column 3: 0 Prime numbers are: 2 3 5 7 === Code Execution Successful ===</pre>

```

main.cpp
38 }
39
40 // Print PRIME numbers
41 cout << "\nPrime numbers are:\n";
42 for (int i = 0; i < rows; i++)
43 {
44     for (int j = 0; j < column; j++)
45     {
46         if (isPrime(arr[i][j]))
47         {
48             cout << arr[i][j] << " ";
49         }
50     }
51     cout << endl;
52 }
53
54 return 0;
55 }

```

Output

```

Enter number of columns: 3
Enter value for row 1 column 1: 2
Enter value for row 1 column 2: 3
Enter value for row 1 column 3: 5
Enter value for row 2 column 1: 4
Enter value for row 2 column 2: 6
Enter value for row 2 column 3: 7
Enter value for row 3 column 1: 9
Enter value for row 3 column 2: 8
Enter value for row 3 column 3: 0

Prime numbers are:
2 3 5
7

```

=== Code Execution Successful ===

Question:3

Java Program Cal.....

Solution:

```

MyJavaFirstProgram.java x Cal.java x
rce History
package myjavafirstprogram;
import java.util.Scanner;
import javafirstclassbsai.Cal;
public class MyJavaFirstProgram {
    public static void main(String[] args)
    {
        Scanner in=new Scanner (System.in);
        Cal c=new Cal ();
        System.out.println("Enter values");
        int a,b,result=5,mul;
        a=in.nextInt();
        b=in.nextInt();
        //c.sum(a,b);
        mul=a*b;
        System.out.println("The sum is" +result+" Multiplication is "+mul)
    }
}

```

put - Java First Class BSAI (run)

run:

Enter values

6

9

The sum is5 Multiplication is 54

BUILD SUCCESSFUL (total time: 6 seconds)

```

package javafirstclassbsai;
public class Cal {
    void sum (int a, int b){
        int res=a+b;
        System.out.println(res);
    }
}

```